

# Diesel oil discrimination by origin and type using physicochemical properties and multivariate analysis

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## ABSTRACT

The objective of this work is to provide models for the prediction and classification of B4 (volume fraction of 4% in conventional diesel) and diesel samples commercialized in the state of Minas Gerais – Brazil, with respect to its origins and types. Hence, we determined physical chemical parameters, such as, specific gravity, flash point and distillation curves for two distinct diesel samples of metropolitan and interior types from different refineries. Through PCA (principal components analysis), the first three principal components separated the samples by origin and types, with percentage of explained variance greater than 96%. For the LDA (linear discriminant analysis), the information in the invoice receipt of samples regarding origin and type were used. In the simultaneous prediction of origin and type, the percentage of correct classification was above 94%, considering the classification error, due to the similarity between some samples.

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## 1. Introduction

Diesel is primarily composed of saturated hydrocarbons (paraffinics), unsaturated (olefins) and aromatic chains that contain between 10 and 19 carbon atoms, which boil in the range of approximately 180–370 °C [1,2]. The process of formulating diesel is acquired by mixing different fractions from the stages of crude oil processing, then the proportions of these components are adjusted to fit the product within standard specifications to ensure optimal performance of engines in general. When processing crude oil to obtain diesel, along with a distillation process, catalytic cracking and hydrocracking can also be employed [3]. Therefore, the difference in chemical composition of crude oil from distinct origins and different refining processes, leads to several types of diesel, with specific characteristics from different refineries [4].

Brazilian gas stations commercialize two types of diesel: metropolitan diesel (MD) and interior diesel (ID). Metropolitan diesel contains around 500 mg kg<sup>-1</sup> of sulfur, which is mandatory to be commercialized within large urban centres, while interior diesel with up to 1800 mg kg<sup>-1</sup> of sulfur, is available in other cities. The ID is marked with a red dye to distinguish it visually from the MD [5], in order to reduce fraud in the commercialization of interior diesel in places where only metropolitan diesel is sold, given that the latter has a higher cost when compared to interior diesel.

In order to obtain diesel with reduced sulfur content, refineries perform the hydrosulfurization process. In this process, organic sulfur compounds are converted in the presence of hydrogen, hydrogen sulfide and sulfur-deficient products, and side reactions may occur depending on the pressure and temperature used [6]. Hydrocracking occurs within higher temperature and pressure, i.e., breaking larger molecules into compounds with lower boiling temperatures. On the other hand, in milder conditions only hydrogenation occurs – process in which unsaturated substances, such as olefins are hydrogenated to produce more stable oils such as paraffin [6]. Thus, depending on the conditions of temperature and pressure used in the hydrosulfurization process, diesel oils with different chemical composition will be obtained. These differences result in distinct physical and chemical properties, such as, distillation and specific gravity, which have specific limits for interior and metropolitan diesel in accordance with the ANP's (National Petroleum Agency) resolution [5].

Currently, both types of diesel must contain 4% of biodiesel, a renewable fuel [7] obtained by the transesterification of animal or vegetable oils, designated as B4. As described in the literature [8], the physicochemical properties of biodiesel are very similar to diesel's properties, and they are miscible in any proportion. Such similarity provides, according to the ANP's resolution [5] to perform quality control tests for the B4 mixture, likewise for pure diesel, following the same specification.

In 2009, roughly 44 billion gallons of diesel were sold in Brazil, including the mix of biodiesel, accounting for 53% of Brazil's automotive fuels market [9]. This large intake is mainly driven by road

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haulage, given that the highway network is the country's main transport mode.

To ensure the quality of the diesel sold, some physical and chemical parameters are monitored using standard procedures (American Society for Testing and Materials – ASTM), and the limits must be met in order for the fuel to be considered in conformity or suitable for consumption. Otherwise, the sample is classified as non-conforming. And non-conformities can be caused by factors such as adulteration, contamination or natural processes. When a sample is in conformity, but has differentiated physicochemical properties, i.e., has a distinct profile from most of the equivalent samples, it is defined as atypical [10].

Surveys conducted by the Laboratory of Fuels of the Federal University of Minas Gerais (LEC-UFMG), using statistical tools [10] and the results of different physicochemical tests, indicated that 20.6% of approximately 3000 diesel samples analyzed in 2009 presented abnormalities in one or more parameters. The atypical behaviour entails two possible causes: planned adulteration with controlled addition of solvents, aiming for the product to stay within specifications and mandatory requirements, or samples with different origins from different refineries [10]. To relate the atypicality of the sample to possible adulteration, initially it is necessary to exclude the possibility that they are from different origins.

Information regarding the fuel's origin can be found in gas stations' purchase receipt, however, they are not always available, and its origin becomes unknown. Fig. 1 presents the origin of almost 3000 samples analyzed by the Fuel Quality Control Monitoring Program (PMQC) of the National Petroleum Agency, Natural Gas and Bio-fuels (ANP), in the eastern portion of Minas Gerais state in 2009, displaying that 56.1% of them do not have an identified origin.

The fuel supplied to the State of Minas Gerais comes from different refineries, and this diversity of sources is more pronounced along its borders. Considering that Brazilian states have different levies on the sale of fuel, it has provided some distributors to unlawfully simulate the sale to a state where taxes are lower, thus engaging into fiscal evasion.

Therefore, tracing the origin of the samples of fuel can be an efficient monitoring mechanism to avoid cases of tax evasion and adulteration practices, providing evidence swiftly against fraud.

Moreover, the possibility of distinguishing the samples of the interior and metropolitan type is relevant, due to the difference in costs of both types of diesel. Fraud can occur by selling interior diesel without dye in areas where only authorized commercialization of metropolitan diesel is allowed, given that the interior diesel has less cost to the distributor, and usually there is no difference in prices for the consumer.

Several studies have been developed in order to determine the origin of extra products of petroleum refining, as motor oil and gasoline, using analytical techniques such as near-infrared [4,11], gas chromatography [12–14] and results of physicochemical assays [15,16] associated with different chemometric tools such as principal component analysis (PCA), linear and regularized discriminant analysis (LDA e RDA), SIMCA (soft independent modeling of class analogy) and MLP (multilayer perceptron).

Balabin and Safieva [11] classified motor oils into three or four classes according to their base stock and kinematic viscosity at low and high temperature. The error of the MLP classification model was half that of RDA method (26% vs. 57%) indicating that MLP is the best method. As the classification errors obtained were high, the authors suggested that the use of more sophisticated classification methods (support vector machines, probabilistic neural networks, etc.) could lead to better classification.

Balabin and Safieva [4] compared the capability of three types of classification – LDA, SIMCA and MLP – associated with near-infrared spectra to distinguish the gasoline samples according to the refinery, refining process and the type (normal regular and premium). Amongst all three of the methods studied, the MLP presented the lowest average error (12%), therefore the most recommended.

Sandercock and Du Pasquier [12–14] used GC/MS–SIM (solid phase extraction techniques and gas chromatography–mass spectrometry with selected ion monitoring) and PCA to analyze samples of gasoline and differentiate them or establish a common origin in cases of forensic investigation of induced fires. The methodology was used and worked perfectly discriminating gasoline samples by origin, retail venues [12] and distinct countries [13], as well as by length of storage [14].

In the prediction of the gasoline's origin Barbeira et al. [15,16] obtained with the use of LDA, percentile close to 97% and 95% for models built from the results of different physicochemical assays

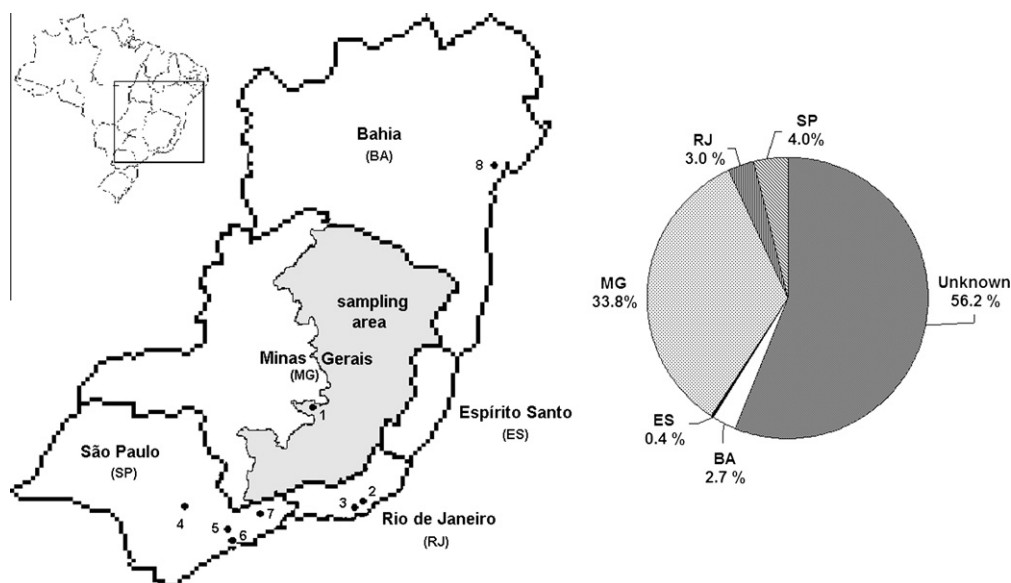


Fig. 1. Sampling area, the location of refineries and origin of the 2838 samples analyzed by the PMQC-ANP, in the eastern section of the state of Minas Gerais in 2008. (1) REGAP; (2) Manguinhos; (3) REDUC; (4) REPLAN; (5) RECAP; (6) RPBC; (7) REVAP and (8) RLAM.

(distillation, specific gravity and infrared spectroscopy) [15] and only using distillation curves [16], respectively.

Considering the significant consumption of diesel by the Brazilian automotive fuel market and the high possibilities of fraud, it is necessary to develop methodologies and analytical tools that contribute to both quality control and the traceability of fuel, either for monitoring tests or forensic investigations.

Therefore, the objective was to provide models for the prediction and classification of B4 and diesel samples with respect to its origins and types (interior and metropolitan), using the results of different physicochemical assays (specific gravity, flash point, and distillation) and distillation curves along with multivariate analysis techniques, such as, LDA and PCA. The lack of studies to determine, simultaneously, the origin and type of diesel reveals the originality of the paper in relation to other works.

## 2. Experimental

A set of 200 samples was classified by origin (refineries), type (metropolitan or interior), and composition (B0 and B4). The refineries origins were identified by the invoice receipt of the purchased products sold by retailers: REGAP, REDUC, RLAM, REVAP and REPLAN. In Table 1, the samples are displayed according to their origin and type.

Each sample was submitted to three physicochemical procedures following the procedures described in the ANP resolution [5], and the results were used for statistical treatment. For specific gravity (ASTM-D4052) [17], flash point (ASTM-D93) [18] and distillation (ASTM-D86) [19] assays at 20 °C, a digital Anton Paar DMA 4500 densimeter, an ISL FP93 flash point analyzer with a closed vessel and Herzog HDA 627 automatic distillers, were used respectively.

The results of the physicochemical assays were arranged into matrixes where the rows corresponded to samples and the columns to the assays results. Two matrixes were built for each set of data. The first matrix was done with results of specific gravity (SG), flash point (FP) and distillation temperatures equivalent to 10%, 50%, 85% and 90% of the recovered volume (T10, T50, T85 and T90, respectively) forming set 1 (C1) and, for the other one, the distillation temperatures were of 4–91% of the recovered volume (T4–T91), at intervals of 1%, forming set 2 (C2). The values for the former four were chosen due to two factors: T50 and T85 have established limits by the Brazilian legislation and because T10 and T90 are used together with specific gravity in the calculation of the cetane number using a formula established by ASTM-D4737 [20].

The data from each matrix were submitted to multivariate analysis, using the PCA method, in order to verify if the diesel samples are separated into groups with respect to the origin and type. In this method, the data are designed in a smaller scale by means of linear combinations of the original variables generating the matrixes of scores and loadings. With the use of PCA, the number of variables is reduced while maintaining a high variance [21]. The LDA was applied after the confirmation by the PCA, of the separation of samples of diesel oil with respect to the origin and type. In order to apply LDA, the information about the origin (refineries), the type (metropolitan/interior), and the addition of biodiesel (B0/B4) was

previously defined, taking into account its general characteristics. For the linear discriminant analysis the whole set of samples was divided randomly into training set (150 samples) and test set, or external validation (50 samples). As internal validation the crossed-validation “leave-one-out” method was used [21], which each sample is left out, once at a time and has its parameter of interest predicted by the new generated model. To assess whether the model had a good categorization of the samples in groups, we used the parameter %CC (percentage of correct classification). This parameter was determined from the total number of samples (XT) and the number of samples correctly classified (XC), from Eq. (1) [21]. Thus, a model with a good categorization has a high %CC, and still, the higher the %CC, the greater the discrimination between samples.

$$\%CC = \frac{X_c * 100}{X_T} \quad (1)$$

The software used to create models of PCA and LDA was Minitab Release (version 14 for Windows).

## 3. Discussion of results

### 3.1. Data arrangement

To build the models using C2, temperatures in the range of 4–91% of the recovered volume were used. According to the ASTM-D86 [19], the samples distillation should occur at the rate of 4–5 mL min<sup>−1</sup>. To achieve this rate, initial settings as resistance heating is required, which can cause vigorous boiling and oscillation of temperature values, from the initial distillation point to approximately 4% v/v. Moreover, in the final stage of distillation, pyrolysis of certain larger molecules can occur, causing a decrease in boiling temperature [19]. As the composition of diesel oil samples differs according to their refineries in origin, not all curves have all points in the final distillation, in the range of 92–99% v/v. Considering these variations, the final and initial points were excluded from the data sets.

The data from SG, FP, T10, T50, T85 and T90 were auto-scaled, in order to give equal importance to all, considering the magnitude of the results were different.

### 3.2. Classification of samples according to origin and type using PCA

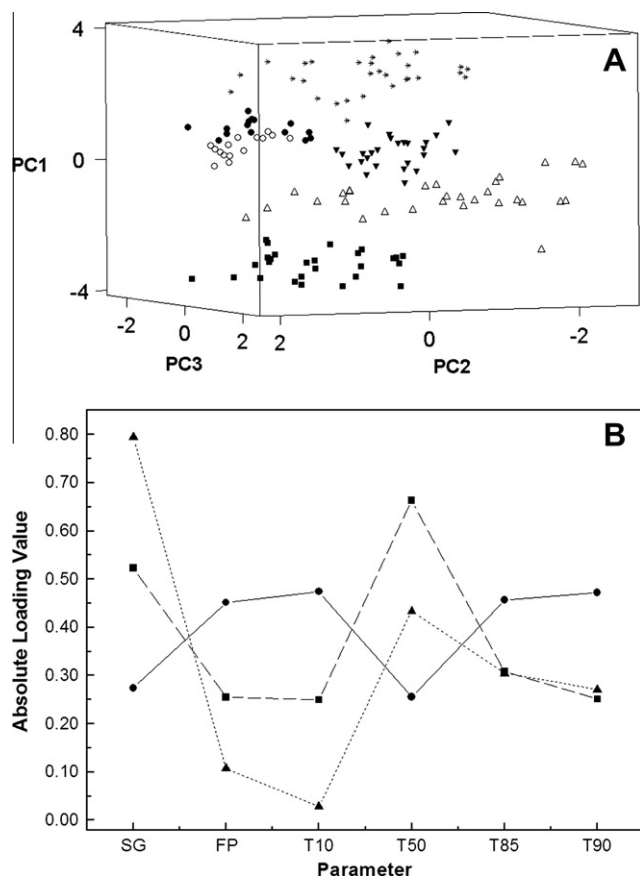
To separate the samples by origins and types, two models were built with samples from C1 (physicochemical assays) and C2 (distillation curves). For these models the explained variance using the first three principal components were equal to 96% (64% PC1, 23.4% PC2 and 8.6% PC3) e 99.1% (61.5% PC1, 36.4% PC2 and 1.2% PC3), respectively. Figs. 2 and 3 present the groups of samples from sets 1 and 2 according to their origin and type, and the loadings of each variable used to form PC1, PC2 e PC3.

The graph of scores using the results of physical and chemical parameters was built from the first three principal components (Fig. 2A). Unlike the graph obtained from C2, it was necessary to use the PC3 to observe a clear separation of the groups formed. This is because the model built from C1 presented an explained variance in PC3 (8.6%) higher than the model obtained using (1.2%).

**Table 1**

Disposition of samples used in classifying samples of diesel oil according to its origin (refineries), type (ID/MD) and the addition of biodiesel (B0/B4).

| Classification | Total of 200 samples |            |             |           |          |
|----------------|----------------------|------------|-------------|-----------|----------|
| Refineries     | 40 REGAP             | 40 RLAM    | 40 REDUC    | 40 REPLAN | 40 REVAP |
| ID or MD       | 21 ID/19 MD          | 40 ID      | 40 ID       | 40 ID     | 40 MD    |
| B0 or B4       | 26 B0/14 B4          | 38 B0/2 B4 | 30 B0/10 B4 | 40 B0     | 40 B0    |

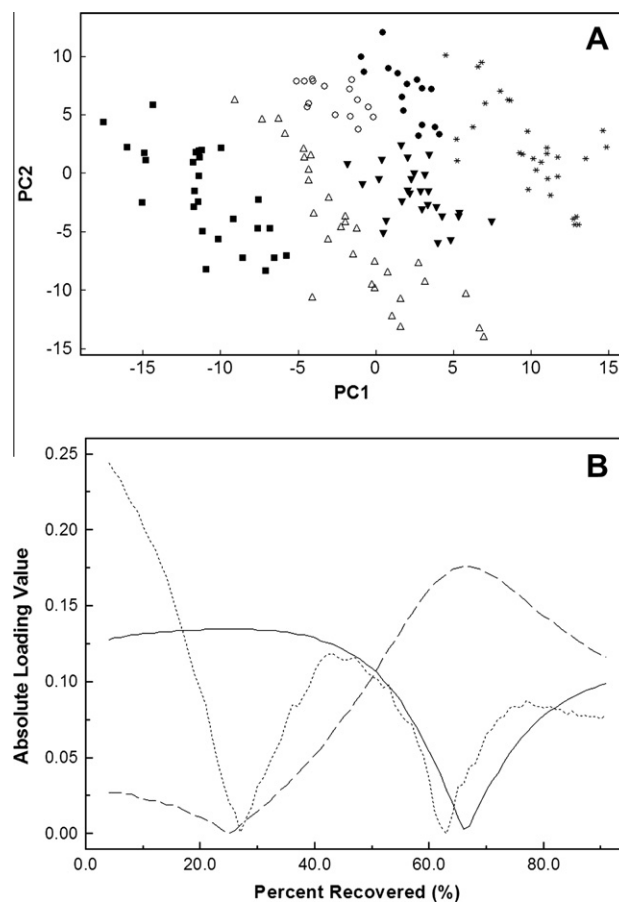


**Fig. 2.** Graph of scores (A) of the first three principal components obtained and the loadings (B) for diesel oil samples analyzed by different physicochemical assays (C1) – formation of groups by origin and type. (○) REGAP-MD; (●) REGAP-ID; (▼) REDUC-ID; (△) RLAM-ID; (■) REPLAN-ID; (✱) REVAP-MD; (—●—) PC1; (—■—) PC2 and (—▲—) PC3.

Analyzing the groups formed from the C1 set (Fig. 2A) showed that the PC1 is responsible for the separation of the samples by origin, except for samples of the metropolitan REGAP type (REGAP-MD) and the REDUC interior type (REDUC-ID), which are separated by PC2. In the loadings graph it is possible to notice that the most important variables for PC2 were specific gravity (SG) and temperature equivalent to 50% of the recovered volume (T50). These variables, in addition to being significant in the creation of PC2, presented different values for REGAP-MD and REDUC-ID groups. One possible explanation for this fact would be the different processes used in petroleum refining to obtain diesel oil, since these processes use raw materials from the same terminal [22].

In the separation by type, only REGAP presented samples of the interior and metropolitan type from the places where it was collected. For these samples, using the C1 set, the combination of the first three principal components was used to visualize the separation of the REGAP-ID and REGAP-MD samples (Fig. 2A). The loadings graph (Fig. 2B) shows that all the variables (physicochemical parameters) were significant to PC1, while the SG and T50 variables were of higher importance to PC2 and PC3.

The separation of samples according to the origin from the C2 set (Fig. 3A) was less clear, only using PC1, when compared to the separation of the model built with the C1 set. For this set, the combination of PC1 and PC2 allowed the segregation of samples by origin, as can be seen in Fig. 3A. The most important variables when building PC1 belongs to the initial region of the distillation (temperatures equivalents to the range of 4–40% of the recovered volume) while for PC2 the final region of the distillation curve



**Fig. 3.** Graph of scores (A) of the first two principal components and loadings (B) obtained for diesel samples analyzed by temperatures equivalent to the percentages of 4–91% of the recovered volume (C2). (○) REGAP-MD; (●) REGAP-ID; (▼) REDUC-ID; (△) RLAM-ID; (■) REPLAN-ID; (✱) REVAP-MD; (—) PC1; (---) PC2 and (---) PC3.

had greater importance (temperatures equivalent to the interval of 60–80% of the recovered volume).

In the classification of samples by type, for C2, the separation of REGAP-ID and REGAP-MD samples was observed using only PC1 (Fig. 3A). Considering that the petroleum used to obtain these samples comes from the same terminal [22], this separation can be attributed to the different refining processes applied. The metropolitan diesel is obtained through the process of hydrodesulfurization, which produces compounds with lower boiling temperatures [6], which accounts for the lower temperatures of the REGAP-MD samples in the initial distillation (Fig. 4).

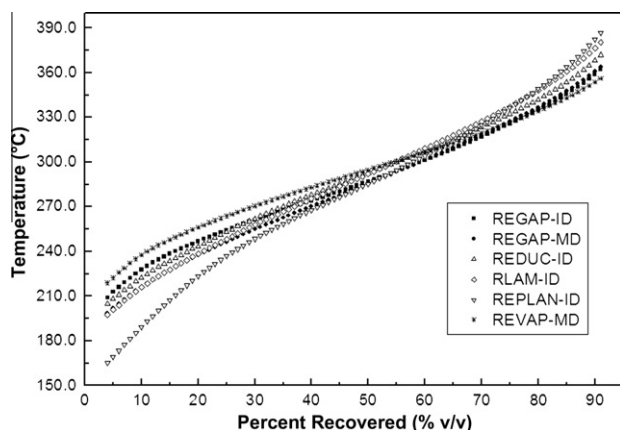
Comparing the models created from C1 and C2, it can be noted that the PCA was an effective tool for classifying samples according to origin and type, using both physicochemical and distillation curves. However, in these models, it was not possible to visualize the separation of the B0 and B4 samples, given that the addition of 4% of biodiesel did not cause significant changes in the physicochemical parameters and distillation curves.

### 3.3. Prediction of origin and type (interior/metropolitan)

LDA models with cross-validation were obtained from samples used in the PCA (sets 1 and 2), given that 150 samples were used in the training set and 50 others in the external validation.

When applying LDA in the training set – obtained from the results of the different physicochemical assays (C1) and distillation curves (C2) – it registered a percentage of correct classification





**Fig. 4.** Average distillation curves of the five refineries used in the study of the prediction of the origin and type of diesel. ( $\Delta$ ) REGAP-MD; ( $\blacktriangle$ ) REGAP-ID; ( $\square$ ) REDUC-ID; ( $\circ$ ) REPLAN-ID; ( $\bullet$ ) RLAM-ID and ( $\nabla$ ) REVAP-MD.

(%CC) of 98% and 95.3%, respectively, in the simultaneous prediction of the origin and type.

While examining Table 2 it shows that the REGAP-ID and REGAP-MD groups have the smallest Mahalanobis distances [23] indicating that these groups presented high similarity with the models, based on physicochemical assays and distillation curves. As seen in the classification of samples by PCA, this similarity is related to the petroleum used to obtain diesel oil by REGAP.

A set of 50 samples was used for external validation, from the results of the physicochemical assays and from the distillation curve. The percentage of correct classification (%CC) obtained in the external validation of C1 and C2 were equal to 96% and 94%, respectively, these percentages being the same as the results obtained in the determination of the origin of the gasoline using LDA associated to different results of physicochemical assays and distillation curves [15,16].

Table 3 C1 presents correct classification of 100%, except for REVAP-MD that presented a percentage of 80%. This decrease in the %CC is due to the misclassification of two REVAP-MD samples that were considered by the model as being a REGAP-ID sample. This misclassification is related to the similarity of the REVAP-MD and REGAP-ID samples that as well as presenting short Mahalanobis distance (Table 2), displayed similar SG and T85 values. While in C2, a decrease in the percentage of accuracy in the prediction of some origins was observed. This fact can be attributed to the similarities between the samples, as evidenced by the similarities of

**Table 3**

Percentages of correct classification (%CC) of the sets of external validation for models built from C1 and C2 in predicting the origin and type of diesel oil samples.

| Origin-type | %CC for C1 | %CC for C2 |
|-------------|------------|------------|
| REGAP-ID    | 100        | 80         |
| REGAP-MD    | 100        | 80         |
| REDUC-ID    | 100        | 90         |
| RLAM-ID     | 100        | 100        |
| REPLAN-ID   | 100        | 100        |
| REVAP-MD    | 80         | 100        |

the equivalent temperatures in the range of 20–40% of the recovered volume for REGAP-ID and REDUC-ID and of 50–91% of the recovered volume for REGAP-ID and REGAP-MD, this is supported by small Mahalanobis distances between groups.

Thus, the use of chemometric tools associated with the outcomes of different physicochemical assays, including distillation curves, permitted the simultaneous prediction of the origin and type of diesel fuel with percentages of correct classification above 94%, regardless of the addition of biodiesel of different oils, which is a positive aspect to the methodology.

### 3.4. Predicting the presence of biodiesel (B0 and B4) in diesel

The linear discriminant analysis was performed using the same set of prediction of origin and type (interior/metropolitan), to predict the presence of biodiesel in diesel. In this set, 28 samples containing 4% of biodiesel (B4) and 172 samples of pure diesel (B0). As noted in Table 1, the small number of B4 samples is related to the fact that only the REGAP, REDUC and RLAM refineries had B4 samples available during the collection period.

The classification of the B0 and B4 samples were done with LDA, even though the formation of groups did not occur when the PCA was applied. The percentage of correct classification for the training sets were 72% and of 79.3% for C1 and C2. For the external validation sets, 70% and 76% of correct classification is obtained for C1 and C2.

In order to verify if the low classification percentages obtained are related to the small number of B4 samples in sets 1 and 2, a model was built with the same quantities of pure diesel oil samples and B4 samples. Alike C1 and C2 the physicochemical parameters, the distillation curves along with the PCA did not cause the segregation of the B0 and B4 samples into different groups. Furthermore, low percentages of correct classification were obtained when this set was submitted to the LDA technique. With these results it was possible to conclude that the models built from C1 and C2 were not biased and that the used techniques (physicochemical parameters and distillation curves) were not able to distinguish samples of B0 and B4. This inefficiency in the classification and prediction of the samples according to the biodiesel content (B0 and B4) is related to the physicochemical similarities of diesel and biodiesel.

## 4. Conclusions

According to the obtained results, it is possible to conclude that the LDA and PCA applied to the results of physicochemical assays (C1), distillation curves (C2) of diesel samples and B4 were adequate tools to classify and predict their origins (several refineries) and type (metropolitan/interior). As the results of sets 1 and 2 were similar, it is clear that both sets can be used to classify and predict origin and type. The distinction between the compositions of (B0 and B4) was not possible to assess using data from C1 and C2.

The use of results of several physicochemical assays or of distillation curves associated with PCA and LDA for the determination of the type of diesel is an excellent advantage, since, in terms of

**Table 2**

Mahalanobis distance between groups in the LDA model with cross-validation and the models built from the physicochemical parameters and distillation curves in the simultaneous prediction of type and origin of diesel oil.

| Origin-type        | C1    | C2     |
|--------------------|-------|--------|
| REDUC-ID/REPLAN-ID | 67.4  | 520.7  |
| REDUC-ID/REVAP-MD  | 31.1  | 264    |
| REDUC-ID/RLAM-ID   | 73    | 107.3  |
| REGAP-ID/REGAP-MD  | 10    | 109.4  |
| REGAP-ID/REDUC-ID  | 31.2  | 144.7  |
| REGAP-ID/REPLAN-ID | 123.5 | 644.2  |
| REGAP-ID/REVAP-MD  | 23    | 206.5  |
| REGAP-ID/RLAM-ID   | 173.1 | 264.2  |
| REGAP-MD/REDUC-ID  | 21.4  | 141.3  |
| REGAP-MD/REVAP-MD  | 34.3  | 212.1  |
| REGAP-MD/REPLAN-ID | 73.7  | 490.6  |
| REGAP-MD/RLAM-ID   | 144.4 | 149.5  |
| REPLAN-ID/REVAP-MD | 171.1 | 1060.7 |
| REPLAN-ID/RLAM-ID  | 119   | 272.7  |
| REVAP-MD/RLAM-ID   | 163.2 | 464.3  |

quality evaluation, only with results of three routine assays, or with distillation, is it possible to determine the origin and type of diesel simultaneously, without the use of complementary assays.

In the case of grouping by composition, it was not possible to segregate the pure diesel samples from the B4 samples, given that low percentages of correct classification in the external validation and training sets were obtained.

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